



COURSE DESCRIPTION FORM: CL-3001: Computer Networks - Lab

INSTITUTION FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad Campus

PROGRAM TO BE EVALUATED BS-CS: Fall-2025
Course Description

Course Code	CL-3001	
Course Title	Computer Networks – Lab	
Credit Hours	1	
Course Instructors	Ms. Maryam Khalid, Ms. Amina Ashfaq, Mr. Hamza Mehmood	
Grading Policy	Absolute	
Policy about missed assessment items in the course	Retake of missed assessment items (other than sessional/ final exam) will not be held. Student who misses an assessment item (other than sessional / final exam) is awarded zero marks in that assessment item i.e. late submission will not be accepted. For missed sessional/ final exam, exam retake/ pretake application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee decide the exam retake/ pretake cases.	
Course Plagiarism Policy	Plagiarism in project or final exam will result in F grade in the lab. Plagiarism in a lab task/assignment will result in zero marks in that particular task/assignment.	
Prerequisites by Course(s) or Topics	CS-2001 Data Structures, CL-2001 Data Structures Lab	
Assessment Instruments with Weights (Sessional exam, final exam, lab Project, lab tasks, etc.)	Assessment with the weight.	
	Assessment Type	Weight
	Lab Tasks (15)	25
	Mid term	15
	Project	20
	Final Exam	40
Lab Coordinator	Ms. Maryam Khalid	
URL (if any)		
Course Catalog Description	Socket programming, TCP architecture, UDP sockets, Network topologies, Static routing, Dynamic routing protocols such as RIP, EIGRP, and OSPF, IP addressing, IP Subnetting, VLSM, IPv4, IPv6, Access lists	
Laboratory Manual	Uploaded on LMS	

Course Goals	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr style="background-color: #d3d3d3;"> <td colspan="3" style="padding: 5px;">A. Course Learning Outcomes (CLOs)</td> </tr> <tr> <td colspan="3" style="padding: 5px;"> 1. Describe utilization of network protocol concepts vis-a-vis OSI and TCP/IP stack. (2) (2) 2. Demonstrate the basics of network concepts using state-of-the-art network tools.(3) (2) 3. Demonstrate operations of various classical routing and switching protocol via simulators.(3) (4) 4. Apply Socket Programming (client/server) to solve various real world problems, including ensuring of data integrity (3) (4) </td> </tr> <tr style="background-color: #d3d3d3;"> <td colspan="3" style="padding: 5px;">B. Program Learning Outcomes (PLOs)</td> </tr> <tr> <td colspan="3" style="padding: 5px;"> For each attribute below, indicate whether this attribute is covered in this course or not. Leave the cell blank if the enablement is little or non-existent. </td> </tr> <tr> <td style="width: 20%; padding: 5px;">1. Academic Education</td><td style="width: 60%; padding: 5px;">Completion of an accredited program of study designed to prepare graduates as computing professionals</td><td style="width: 20%; text-align: center; padding: 5px;">✓</td></tr> <tr> <td style="padding: 5px;">2. Knowledge for Solving Computing Problems</td><td style="padding: 5px;">Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems.</td><td style="text-align: center; padding: 5px;">✓</td></tr> <tr> <td style="padding: 5px;">3. Problem Analysis</td><td style="padding: 5px;">Identify, formulate, research literature, and analyze complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.</td><td style="text-align: center; padding: 5px;">✓</td></tr> <tr> <td style="padding: 5px;">4. Design/ Development of Solutions</td><td style="padding: 5px;">Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations</td><td style="text-align: center; padding: 5px;">✓</td></tr> <tr> <td style="padding: 5px;">5. Modern Tool Usage</td><td style="padding: 5px;">Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations</td><td style="text-align: center; padding: 5px;">✓</td></tr> <tr> <td style="padding: 5px;">6. Individual and Team Work</td><td style="padding: 5px;">Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings</td><td style="text-align: center; padding: 5px;">✓</td></tr> <tr> <td style="padding: 5px;">7. Communication</td><td style="padding: 5px;">Communicate effectively with the computing community and with society at large about</td><td style="text-align: center; padding: 5px;">✓</td></tr> </table>	A. Course Learning Outcomes (CLOs)			1. Describe utilization of network protocol concepts vis-a-vis OSI and TCP/IP stack. (2) (2) 2. 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		complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions	
	8. Computing Professionalism and Society	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice	
	9. Ethics	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice	
	10. Life-long Learning	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional	

C. Mapping of CLOs on PLOs (CLO: Course Learning Outcome, PLOs: Program Learning)											
		PLOs									
		1	2	3	4	5	6	7	8	9	10
C L O s	1		✓								
	2		✓								
	3				✓						
	4				✓						

Topics covered in the course (assume 15-week instruction and 3 contact hours per week)	1. Topics to be covered:																								
	<table><tr><th>List of Topics</th><th>No. of Weeks</th><th>Contact Hours</th><th>CLO(s)</th></tr><tr><td>Introduction to Packet Tracer and Network Topology Setup</td><td>1</td><td>3</td><td>1,2</td></tr><tr><td>Protocol Layer Simulation and Wireless connections in Packet Tracer</td><td>1</td><td>3</td><td>1,2</td></tr><tr><td>Capturing HTTP/HTTPS packets using Wireshark</td><td>1</td><td>3</td><td>1,2</td></tr><tr><td>Configuring DNS and Email Servers in Packet Tracer</td><td>1</td><td>3</td><td>1,2</td></tr><tr><td>Socket Programming in C/C++ TCP Chat Application</td><td>1</td><td>3</td><td>4</td></tr></table>	List of Topics	No. of Weeks	Contact Hours	CLO(s)	Introduction to Packet Tracer and Network Topology Setup	1	3	1,2	Protocol Layer Simulation and Wireless connections in Packet Tracer	1	3	1,2	Capturing HTTP/HTTPS packets using Wireshark	1	3	1,2	Configuring DNS and Email Servers in Packet Tracer	1	3	1,2	Socket Programming in C/C++ TCP Chat Application	1	3	4
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	Socket Programming in C/C++ – UDP Chat Application	1	3	4
	Concurrency in socket programming + File Transfer over TCP/Simulated Packet loss in UDP	1	3	4
	Wireshark TCP Flow Analysis (3-Way Handshake, RTT, Window Size)	1	3	1,2
	IP Addressing and Subnetting Practice in Packet Tracer + static routing	1	3	1,3
	Routing Protocol Simulation RIP	1	3	3
	Routing Protocol Simulation OSPF and EIGRP in Packet Tracer	1	3	3
	Subnetting and VLSM	1	3	1,3
	Configuring DHCP and NAT in Packet Tracer	1	3	2,3
	BGP Simulation in Cisco PT & ICMP Analysis with Wireshark	1	3	1,2,3
	VLAN, ACL Configuration and Error Detection (CRC) in Packet Tracer	1	3	2,3
	Total	15	45	
Practical/ Programming Work/ Tools	Cisco Packet Tracer 8.2, Wireshark 4.4, WSL(Windows subsystem for Linux), gcc and g++			
Lab Time Spent (in percentage)	Theory	Problem Analysis	Solution Design	Social and Ethical Issues
	5	15	75	5